new analysis: MOTHUR version 1.48.30

Make sure you know the number of samples you are processing. In my case for Tegu Project(93 samples & 2 extraction blanks & 1 pcr blank/192 files)
Note: Preemptively give your samples short names, Mothur has a hard time recognizing longer names.

Project Directory in ISAAC:
/lustre/isaac24/proj/UTK0268/UTKGenomicsCoreSequencingData/250319_VL00838_18_AAGLWYYM5/UTK0268_DeBruyn_amplicon_250319

Make a master folder where all your files will go for bioinformatics processing:
Tegu_ITS2

In the Tegu_ITS2 folder:

- FASTQ_Generation_TeguITS2 file folder *(this is the files downloaded off of the cluster)*
- FASTQ_TeguITS2_Extracted *(extracted .gz to .fastq files, this is the 3rd file folder that was made)*
- TEGUITS2.fasta *(**__this is the beginning working file folder__ that has all of the extracted .fasta files, with their filenames changed to whatever it is I want to see on the outputs*)
- Mothur.exe

#I had to do these steps on my personal laptop as my drive would not work with 7zip on lab computer
In ISAAC cluster:

- go to email with file pathway
- open on demand
- files > home directory
- change directory, copy paste
- download necessary files to hard drive (.fastq.gz)

Extract .gz files into new folder using 7-zip into the fastq_extracted folder

- unzips the files, but still need to be opened (go into the file, put * in the search bar, and copy the opened files to the working folder)
- also copy and paste the uchime and vchime files into the working folder

Now that the files are set up, we're ready to begin the mothur pipeline.
Double click on mothur.exe (located in the Tegu_ITS2 folder) and run the following:
mothur > make.file(inputdir=TEGUITS2.fasta, type=fastq, prefix=TEGUITS2)

Here's the output from this command (below). This took a couple of minutes.

```
Setting input directories to:
    TEGUITS2.fasta\
```

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Output File Names:
TEGUITS2.fasta\TEGUITS2.files

Note: Typically, one ".files" file will be made. However, sometimes mothur makes two files, the ".single.files" and the ".paired.files". Any files that have names that don't match go into the single files and the paired files have matching names. Check in notepad if this occurs.

For ease of use move the mothur.exe file into TEGUITS2.fasta and work from within the file.
Now we merge paired reads with the following command:
mothur> make.contigs(file= TEGUITS2.files, processors=8)
This took about **2 hours**. It created the following output files (below) located in the TEGUITS2.fasta file folder.

Total of all groups is 23929585

It took 7677 secs to process 23929585 sequences.

Output File Names:
TEGUITS2.trim.contigs.fasta
TEGUITS2.scrap.contigs.fasta
TEGUITS2.contigs_report
TEGUITS2.contigs.count_table

**NOTE: double check to make sure all of your samples are present.*

Group count:
BLANK1 16505
BLANK2 6980
PCRBlank 6879
T0C 556509
T0SB 361412
T0T 401585
T10C 293594
T10SB 74556
T10T 309590
T11C 536646
T11SB 174773
T11T 412044
T13C 393832
T13SB 27173
T13T 403348
T14C 509425
T14SB 89759
T14T 408875

T15C 39768
T15SB 105320
T15T 276463
T16C 313763
T16SB 323027
T16T 334790
T17C 457214
T17SB 143975
T17T 374984
T18C 75745
T18SB 26558
T18T 41563
T19C 291419
T19SB 10847
T19T 181488
T1C 275712
T1SB 442908
T1T 222264
T20C 412530
T20SB 70050
T20T 421021
T21C 91422
T21SB 251907
T21T 26844
T22C 159399
T22SB 4568
T22T 51134
T23C 432405
T23SB 384805
T23T 350767
T24C 306963
T24SB 408733
T24T 205468
T25C 324043
T25SB 375664
T25T 341562
T26C 47865
T26SB 424832
T26T 325832
T27C 254634
T27SB 409478
T27T 265736
T28C 186792
T28SB 230266
T28T 419408

T29C 105933
T29SB 147819
T29T 394390
T2C 62230
T2SB 201885
T2T 170031
T30C 169799
T30SB 497283
T30T 348227
T31C 97022
T31SB 398186
T31T 106270
T3C 110117
T3SB 33003
T3T 254137
T4C 294761
T4SB 38618
T4T 383309
T5C 733568
T5SB 24984
T5T 215941
T6C 503052
T6SB 133936
T6T 276912
T7C 204032
T7SB 387640
T7T 476259
T8C 226209
T8SB 430134
T8T 95997
T9C 72355
T9SB 110339
T9T 119786

running the following command takes a look at what we have so far:

mothur> summary.seqs(fasta= TEGUITS2.trim.contigs.fasta) provides the following info:

Using 8 processors.

	Start	End	NBases	Ambigs	Polymer	NumSeqs
Minimum:	1	300	300	0	2	1
2.5%-tile:	1	308	308	0	4	598240
25%-tile:	1	339	339	0	4	5982397
Median:	1	362	362	0	5	11964793
75%-tile:	1	404	404	0	6	17947189
97.5%-tile:	1	583	583	3	210	23331346
Maximum:	1	600	600	128	300	23929585

Mean: 1 383 383 0 17
of Seqs: 23929585

It took 324 secs to summarize 23929585 sequences.

Output File Names:
TEGUITS2.trim.contigs.summary

If you know what you're doing, you can run the make.contigs command (above) and remove the primers at the same time [make.contigs (file=filename, oligos=oligofile.txt, processors=8)]. We only made the contigs (assembled the forward and reverse reads) and left the primers attached. There are several commands available to remove primers, one of which is pcr.seqs

A file containing primer sequences was inserted into the TEGUITS2.fasta file. Mothur typically refers to primers and other adapters etc as oligos and refers to them in that manner in their documentation. Oligos files are set up as text (.txt) files—it is easiest to do this is a text editor such as notepad. In this case the oligo file is named Creggerprimers2.txt and the contents are as follows:

forward CATCGATGAAGAACGCAG
forward CAACGATGAAGAACGCAG
forward CACCGATGAAGAACGCAG
forward CATCGATGAAGAACGTAG
forward CATCGATGAAGAACGTGG
forward CATCGATGAAGAACGCTG
reverse TCCTSCGCTTATTGATATGC
reverse TCCTCGCCTTATTGATATGC

The following command was run in order to trim the various primers off (run time took ~50 hours, this varies highly on number of sequences and computing power):

mothur> pcr.seqs(fasta=TEGUITS2.trim.contigs.fasta, oligos=Creggerprimers2.txt, pdiffs=2, rdiffs=2)

Output File Names:
TEGUITS2.trim.contigs.pcr.fasta
TEGUITS2.trim.contigs.bad.accnos
TEGUITS2.trim.contigs.scrap.pcr.fasta
TEGUITS2.contigs.pcr.fasta

running the following command takes a look at what we have so far:

mothur> summary.seqs(fasta=TEGUITS2.trim.contigs.pcr.fasta)

Using 16 processors.

	Start	End	NBases	Ambigs	Polymer	NumSeqs
Minimum:	1	1	1	0	1	1
2.5%-tile:	1	195	195	0	4	493606
25%-tile:	1	296	296	0	4	4936054
Median:	1	318	318	0	5	9872107
75%-tile:	1	359	359	0	6	14808160
97.5%-tile:	1	430	430	2	9	19250607
Maximum:	1	561	561	124	217	19744212
Mean:	1	322	322	0	5	
# of Seqs:	19744212					

It took 150 secs to summarize 19744212 sequences.

Output File Names:
TEGUITS2.trim.contigs.pcr.summary

#running command: *(*Note: this remove.seqs step before precluster is new to the pipeline, but it is necessary in order for the precluster command to run)*
remove.seqs(accnos=TEGUITS2.trim.contigs.bad.accnos,count= TEGUITS2.contigs.count_table)

Removed 4185373 sequences from TEGUITS2.contigs.count_table.

Output File Names:
TEGUITS2.contigs.pick.count_table

#THIS STEP IS OPTIONAL AND MAY BE HELPFUL TO YOU FOR ORGANIZATION:
You may want to create a separate quality control file. This will contain your current .count_table and current .fasta files, along with a copy of the mothur and vsearch executable (mothur.exe, vsearch.exe) files. Running mothur from within a file will help cut down on the long file path names…if not, mothur can be called from a directory up and filepath specified as has been done previously.

The next command will ordinarily be used to remove sequences with ANY ambiguous bases and sequences less than 50 and greater than 275 in the case of 16S processing. According to Cregger (2018), only reads below 200 bp were removed. The ITS2 region that we are amplifying here is about 300-400 bp, and fungal ITS regions can vary considerably in size. While it is possible that there may be long sequences with minimal overlap, it is worth considering not removing long sequences in order not to lose information at the top end. So while the general command may read as follows: screen.seqs(fasta=TEGUITS2.trim.contigs.pcr.fasta, count=TEGUITS2.contigs.count_table, maxambig=0, maxlength=275)….we instead run an edited version (immediately below).

The following command was run (either from in the TEGUITS2.fasta file or the seperate QC file if you made one as described above) This takes about 10 minutes:

screen.seqs(fasta=TEGUITS2.trim.contigs.pcr.fasta, count=TEGUITS2.contigs.pick.count_table, maxambig=0, minlength=200)

It took 108 secs to screen 19744212 sequences, removed 4207868.

/**/

Running command: remove.seqs(accnos=TEGUITS2.trim.contigs.pcr.bad.accnos.temp, count=TEGUITS2.contigs.pick.count_table)

Removed 4207868 sequences from TEGUITS2.contigs.pick.count_table.

Output File Names:

TEGUITS2.contigs.pick.pick.count_table

/**/

Output File Names:

TEGUITS2.trim.contigs.pcr.good.fasta

TEGUITS2.trim.contigs.pcr.bad.accnos

TEGUITS2.contigs.pick.good.count_table

It took 519 secs to screen 19744212 sequences.

running the following command takes a look at what we have so far:

summary.seqs(fasta=TEGUITS2.trim.contigs.pcr.good.fasta)

Using 16 processors.

	Start	End	NBases	Ambigs	Polymer	NumSeqs
Minimum:	1	200	200	0	3	1
2.5%-tile:	1	201	201	0	4	388409
25%-tile:	1	297	297	0	4	3884087
Median:	1	316	316	0	5	7768173
75%-tile:	1	357	357	0	6	11652259
97.5%-tile:	1	422	422	0	8	15147936
Maximum:	1	560	560	0	217	15536344
Mean:	1	324	324	0	5	
# of Seqs:	15536344					

It took 98 secs to summarize 15536344 sequences.

Output File Names:
TEGUITS2.trim.contigs.pcr.good.summary

Note: *we originally had 23,929,585 sequences….pcr.seqs removed 4,185,373, and screen.seq (removing ambiguous bases removed 4,207,868. At the moment we have 15,536,344, which is about 65% of what we started with.*

The following command was run (from within your working folder). This will keep only the unique sequences:
unique.seqs(fasta=TEGUITS2.trim.contigs.pcr.good.fasta,count=TEGUITS2.contigs.pick.good.count_table
)

Output File Names:
TEGUITS2.trim.contigs.pcr.good.unique.fasta
TEGUITS2.trim.contigs.pcr.good.count_table
**Note: we used to rely on .names files in the older pipelines, but 1.48 and onward focus on count files, which are smaller and more efficient*.

running the following command takes a look at what we have so far:
mothur > summary.seqs(fasta=TEGUITS2.trim.contigs.pcr.good.unique.fasta,
count=TEGUITS2.trim.contigs.pcr.good.count_table)

Using 16 processors.

	Start	End	NBases	Ambigs	Polymer	NumSeqs
Minimum:	1	200	200	0	3	1
2.5%-tile:	1	201	201	0	4	388409
25%-tile:	1	297	297	0	4	3884087
Median:	1	316	316	0	5	7768173
75%-tile:	1	357	357	0	6	11652259
97.5%-tile:	1	422	422	0	8	15147936
Maximum:	1	560	560	0	217	15536344
Mean:	1	324	324	0	5	
# of unique seqs:	874466					
total # of seqs:	15536344					

It took 19 secs to summarize 15536344 sequences.

Output File Names:
TEGUITS2.trim.contigs.pcr.good.unique.summary

Note: *we originally had 23,929,585 sequences….pcr.seqs removed 4,185,373, and screen.seq (removing ambiguous bases) removed 4,207,868 leaving 15,536,344. Running unique.seqs reduced this to 874,466 unique sequences.*

#We ran the following command (and it took almost 24 hours!!). **Note: I've had some hiccups with this command. Typically you would want to run it with 12 processors, but I reduced it to 8 because otherwise it would make mothur crash. This may be dependent on the computer used.*

pre.cluster(fasta=TEGUITS2.trim.contigs.pcr.good.unique.fasta,count=TEGUITS2.trim.contigs.pcr.good.count_table,diffs=3,processors=8)

It took 1 secs to merge 262428 sequences group data.
/**/
Running get.seqs:
Selected 175446 sequences from TEGUITS2.trim.contigs.pcr.good.unique.fasta.
/**/
It took 63275 secs to run pre.cluster.

Using 8 processors.

Output File Names:
TEGUITS2.trim.contigs.pcr.good.unique.precluster.fasta
TEGUITS2.trim.contigs.pcr.good.unique.precluster.count_table
TEGUITS2.trim.contigs.pcr.good.unique.precluster.BLANK1.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T0C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T0SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T0T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T10C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T10SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T10T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T11C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T11SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T11T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T13C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T13SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T13T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T14C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T14SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T14T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T15C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T15SB.map

TEGUITS2.trim.contigs.pcr.good.unique.precluster.T15T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T16C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T16SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T16T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T17C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T17SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T17T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T18C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T18SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T18T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T19C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T19SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T19T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T1C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T1SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T1T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T20C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T20SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T20T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T21C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T21SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T21T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T22C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T22SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T22T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T23C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T23SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T23T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T24C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T24SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T24T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T25C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T25SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T25T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T26C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T26SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T26T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T27C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T27SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T27T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T28C.map

TEGUITS2.trim.contigs.pcr.good.unique.precluster.T28SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T28T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T29C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T29SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T29T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T2C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T2SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T2T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T30C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T30SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T30T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T31C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T31SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T31T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T3C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T3SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T3T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T4C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T4SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T4T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T5C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T5SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T5T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T6C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T6SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T6T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T7C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T7SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T7T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T8C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T8SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T8T.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T9C.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T9SB.map
TEGUITS2.trim.contigs.pcr.good.unique.precluster.T9T.map

#Use this command to summarize the data:

mothur > summary.seqs(fasta=TEGUITS2.trim.contigs.pcr.good.unique.precluster.fasta, count=TEGUITS2.trim.contigs.pcr.good.unique.precluster.count_table)

Using 8 processors.

	Start	End	NBases	Ambigs	Polymer	NumSeqs
Minimum:	1	200	200	0	3	1
2.5%-tile:	1	200	200	0	4	388409
25%-tile:	1	297	297	0	4	3884087
Median:	1	317	317	0	5	7768173
75%-tile:	1	357	357	0	6	11652259
97.5%-tile:	1	422	422	0	8	15147936
Maximum:	1	560	560	0	217	15536344
Mean:	1	324	324	0	5	

of unique seqs: 175446
total # of seqs: 15536344

It took 1 secs to summarize 15536344 sequences.

Output File Names:
TEGUITS2.trim.contigs.pcr.good.unique.precluster.summary

#We ran the following command (it took about 10 minutes)
mothur> chimera.vsearch(fasta=TEGUITS2.trim.contigs.pcr.good.unique.precluster.fasta, count=TEGUITS2.trim.contigs.pcr.good.unique.precluster.count_table,dereplicate=t)

Removing chimeras from your input files:
/**/
Running command: remove.seqs(fasta=TEGUITS2.trim.contigs.pcr.good.unique.precluster.fasta, accnos=TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.accnos)
Removed 52997 sequences from TEGUITS2.trim.contigs.pcr.good.unique.precluster.fasta.

Output File Names:
TEGUITS2.trim.contigs.pcr.good.unique.precluster.pick.fasta

/**/

Output File Names:
TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.count_table
TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.chimeras
TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.accnos
TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.fasta

#We ran the following command:
mothur > remove.seqs(fasta=current, accnos=current)

Using TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.accnos as input file for the accnos parameter.
Using TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.fasta as input file for the fasta parameter.
Removed 0 sequences from
TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.fasta.

Output File Names:
TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.fasta

NOTE : the command to remove the accnos may not be necessary in the new version of mothur

#We ran the following command:

mothur > summary.seqs(fasta=current, count=current)

Using TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.count_table as input file for the count parameter.
Using TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.fasta as input file for the fasta parameter.

Using 16 processors.

	Start	End	NBases	Ambigs	Polymer	NumSeqs
Minimum:	1	200	200	0	3	1
2.5%-tile:	1	200	200	0	4	384263
25%-tile:	1	297	297	0	4	3842621
Median:	1	318	318	0	5	7685241
75%-tile:	1	357	357	0	6	11527861
97.5%-tile:	1	422	422	0	8	14986219
Maximum:	1	560	560	0	217	15370481
Mean:	1	324	324	0	5	
# of unique seqs:	122449					
total # of seqs:	15370481					

It took 1 secs to summarize 15370481 sequences.

Output File Names:
TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.summary

Download UNITE database release 10.0 (2025-02-19) for fungi. You will have to scroll down to find it. Remember that the version of the UNITE data base that you use is dependent on the version of Mothur that you are using. This part was deceptively hard, some of the new fungal versions only provide two files, a fasta and a txt file, this version below is the correct version to use!

There are full "UNITE+INSD" datasets available in mothur format:

When using this resource, please cite it as follows:
Abarenkov, Kessy; Zirk, Allan; Piirmann, Timo; Pöhönen, Raivo; Ivanov, Filipp; Nilsson, R. Henrik; Kõljalg, Urmas (2025): UNITE mothur release for Fungi. Version 19.02.2025. UNITE Community.
https://doi.org/10.15156/BIO/3301233

Includes singletons set as RefS (in dynamic files).

Steps for downloading the UNITE database—this is actually a bit tricky!
1. Go to unite.ut.ee, click the resources tab and go to the Mothur release lower down on the page.
2. Click the DOI link within the Mothur release (10.0 in this case)—you are looking for the dynamic files which are curated.
3.At the bottom of the page there is a zip file (sh_mothur_release_2025-02-19.tgz)with the download for this release, click it and this will begin the download. Save this to your passport in the project file (You may be prompted to add your email and explain your purpose for downloading)
4. Once the download is complete extract the file.
5. NOW you should have a folder (sh_mothur_release_04.02.2020) that contains six files: UNITEv10_sh_97.fasta, UNITEv10_sh_97.tax, UNITEv10_sh_99.fasta, UNITEv10_sh_99.tax, UNITEv10_sh_dynamic.fasta, UNITEv10_sh_dynamic.tax
7) You will only need one pair of files (.fasta and .tax). For our purposes we are using the dynamic files. Copy these into the working folder.

#We ran the following command (it took about 1 hour)
The original command is as follows:
mothur
>classify.seqs(fasta=TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.fasta, count= TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.count_table, reference=UNITEv10_sh_dynamic.fasta, taxonomy= UNITEv10_sh_dynamic.tax)

WARNING]: mothur reversed some your sequences for a better classification. If you would like to take a closer look, please check TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.UNITEv10_sh_dynamic.wang.flip.accnos for the list of the sequences.

It took 3316 secs to classify 122449 sequences.

It took 8 secs to create the summary file for 122449 sequences.

Output File Names:
TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.UNITEv10_sh_dynamic.wang.taxonomy

TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.UNITEv10_sh_dynamic.wang.tax.summary
TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.UNITEv10_sh_dynamic.wang.flip.accnos

#We ran the following command:
mothur > remove.lineage(fasta=current, count=current, taxonomy=TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.UNITEv10_sh_dynamic.wang.taxonomy, taxon=unknown)

Using TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.count_table as input file for the count parameter.
Using TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.fasta as input file for the fasta parameter.

*** No contaminants to remove ***

Output File Names:
TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.UNITEv10_sh_dynamic.wang.pick.taxonomy

**Note: in the pipeline, there should be 3 output files from remove.lineage, and I only have one, but then again it said nothing to remove. So, all following commands will take a mixture of output files.*

We move on to clustering sequences with greedy clustering, using a cutoff of 0.05 and method=agc (this is a VSEARCH clustering option because contigs are of differing lengths, therefore no dist matrix can be calculated)

#We ran the following command (~5 minutes):
cluster(fasta=current, count=current, method=agc, cutoff=0.05)

Using TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.count_table as input file for the count parameter.
Using TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.fasta as input file for the fasta parameter.

Using 16 processors.

0.05
vsearch v2.16.0_win_x86_64, 31.9GB RAM, 16 cores
https://github.com/torognes/vsearch

Reading file
TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.fasta.sorted.fasta.temp
100%
47554629 nt in 122449 seqs, min 200, max 560, avg 388
Masking 100%
Counting k-mers 100%
Clustering 100%
Sorting clusters 100%
Writing clusters 100%
Clusters: 16130 Size min 1, max 3814, avg 7.6
Singletons: 8683, 7.1% of seqs, 53.8% of clusters
It took 131 seconds to cluster

Output File Names:
TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.agc.list

Make the OTU table.
#We ran the following command:
make.shared(list=TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.agc.list, count=current)

Using TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.count_table as input file for the count parameter.
0.05

Output File Names:
TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.agc.shared

And now classify the OTUs based UNITE database taxonomy
#We ran the following command:
classify.otu(list=TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.agc.list, count=current,taxonomy=TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.UNITEv10_sh_dynamic.wang.pick.taxonomy)

Using TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.count_table as input file for the count parameter.
0.05

Output File Names:
TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.agc.0.05.cons.taxonomy
TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.agc.0.05.cons.tax.summary

Shared (from make.shared) and Taxonomy (from classify.otu) are now used for downstream processing in Phyloseq.

Shared: TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.agc.shared

Taxonomy: TEGUITS2.trim.contigs.pcr.good.unique.precluster.denovo.vsearch.pick.agc.0.05.cons.taxonomy

#Note: it is probably easiest to change these names for downstream analysis